

TE Curve Verification Pipeline Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This update accepts the completed `wave 3.3` curve-aware training campaign into the official `TE Curve Verification Pipeline` offline verification matrix.

Decision

- `wave 3.3` is verified as an exploratory curve-aware training baseline.
- No `wave 3.3` candidate is promoted over the current accepted `TE Curve Verification Pipeline` leaders.
- The project continues to maintain three parallel best surfaces: `Fw` , `Bw` , and `global` .
- Within `wave 3.3` , the strongest forward candidate is `track2g_curve_aware_raw_centered_shape_Fw` .
- Within `wave 3.3` , the strongest backward candidate is `track2g_curve_aware_pointwise_control_Bw` .
- Within `wave 3.3` , the strongest global candidate is `track2g_curve_aware_full_curve_composite_global` .

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:
doc/reports/analysis/track2/Track 2 Directional Model Comparison.md ;
- matrix summary:
output/validation_checks/track2_reference_comparison/2026-06-09-13-25-56__track2_full_directional_family_matrix_track2g_curve_aware_track2_refresh_2026_06_09/validation_summary.yaml ;
- per-condition metrics:
output/validation_checks/track2_reference_comparison/2026-06-09-13-25-56__track2_full_directional_family_matrix_track2g_curve_aware_track2_refresh_2026_06_09/per_condition_metrics.csv ;
- validation report:
doc/reports/analysis/validation_checks/track2/2026-06-09-13-33-45_track2_full_directional_family_matrix_track2g_curve_aware_track2_refresh_2026_06_09_report.md ;
- best-model collage report:
doc/reports/analysis/track2/best_model_collage_report/[2026-06-09]/track2_best_model_collage_report.md ;
- multi-model curve comparison report:
doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-06-09]/track2_multi_model_curve_comparison_report.md .

Candidate Refresh

The refresh added 12 registry-backed Wave 3.3 candidates. Because the baseline summary used by the operator launcher predated the Wave 3.1 and Wave 3.2 refreshes, the incremental package also rechecked the existing 3 Wave 3.1 and 6 Wave 3.2 candidates. The official decision scope of this report is the Wave 3.3 candidate family.

Surface	Loss Profile	Candidate	Registry Family
global	pointwise control	track2g_curve_aware_pointwise_control_global	track2g_curve_aware_harmonic_residual_offset_pointwise_control_global
Fw	pointwise control	track2g_curve_aware_pointwise_control_Fw	track2g_curve_aware_harmonic_residual_offset_pointwise_control_fw
Bw	pointwise control	track2g_curve_aware_pointwise_control_Bw	track2g_curve_aware_harmonic_residual_offset_pointwise_control_bw
global	raw plus centered shape	track2g_curve_aware_raw_centered_shape_global	track2g_curve_aware_harmonic_residual_offset_raw_centered_shape_global
Fw	raw plus centered shape	track2g_curve_aware_raw_centered_shape_Fw	track2g_curve_aware_harmonic_residual_offset_raw_centered_shape_fw

Surface	Loss Profile	Candidate	Registry Family
Bw	raw plus centered shape	track2g_curve_aware_raw_centered_shape_Bw	track2g_curve_aware_harmonic_residual_offset_raw_centered_shape_bw
global	raw plus offset	track2g_curve_aware_raw_offset_global	track2g_curve_aware_harmonic_residual_offset_raw_offset_global
Fw	raw plus offset	track2g_curve_aware_raw_offset_Fw	track2g_curve_aware_harmonic_residual_offset_raw_offset_fw
Bw	raw plus offset	track2g_curve_aware_raw_offset_Bw	track2g_curve_aware_harmonic_residual_offset_raw_offset_bw
global	full curve composite	track2g_curve_aware_full_curve_composite_global	track2g_curve_aware_harmonic_residual_offset_full_curve_composite_global
Fw	full curve composite	track2g_curve_aware_full_curve_composite_Fw	track2g_curve_aware_harmonic_residual_offset_full_curve_composite_fw
Bw	full curve composite	track2g_curve_aware_full_curve_composite_Bw	track2g_curve_aware_harmonic_residual_offset_full_curve_composite_bw

The matrix now contains 132 candidates.

Verification Rules

Surface	Training or Archive Scope	Evaluation Scope
Fw	forward-only training or archive	forward TE Curve Verification Pipeline curves only
Bw	backward-only training or archive	backward TE Curve Verification Pipeline curves only
global	forward and backward training together	both directions, reported by direction and combined

The global , Fw , and Bw branches are carried forward in parallel. They are not collapsed into one scalar winner.

Current Leaders

Best Composite References

Candidate	Source	Direction	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
paper_retuned_best_Fw	rcim_retuned	forward	0.001839	0.002041	4.109	9.866
paper_retuned_best_Bw	rcim_retuned	backward	0.003675	0.004284	7.572	15.645

Repository-Owned Static Baselines

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
forward	tree_global	0.002998	0.003364	6.590	11.601
backward	tree_Bw	0.003258	0.003651	7.051	14.116
global combined	tree_global	0.003144	0.003533	6.854	13.314

Repository-Owned Neural Leaders

Surface	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
Fw	periodic_gru_sequence_Fw	0.003186	0.003438	7.077	13.323
Bw	periodic_gru_sequence_Bw	0.002392	0.002639	5.466	14.820
global combined	periodic_gru_sequence_global	0.002704	0.002949	6.139	13.396

Wave 3.3 Result

Surface	Candidate	Loss Profile	MAE [deg]	RMSE [deg]	Mean [%]
Fw	track2g_curve_aware_raw_centered_shape_Fw	raw plus centered shape	0.003174	0.003429	7.047
Fw	track2g_curve_aware_full_curve_composite_Fw	full curve composite	0.003251	0.003515	7.209
Fw	track2g_curve_aware_raw_offset_Fw	raw plus offset	0.003269	0.003588	7.268
Fw	track2g_curve_aware_pointwise_control_Fw	pointwise control	0.003362	0.003612	7.474
Bw	track2g_curve_aware_pointwise_control_Bw	pointwise control	0.003436	0.003761	7.538
Bw	track2g_curve_aware_raw_centered_shape_Bw	raw plus centered shape	0.003465	0.003790	7.582
Bw	track2g_curve_aware_raw_offset_Bw	raw plus offset	0.003469	0.003799	7.608
Bw	track2g_curve_aware_full_curve_composite_Bw	full curve composite	0.003510	0.003897	7.683
global combined	track2g_curve_aware_full_curve_composite_global	full curve composite	0.003338	0.003649	7.364

Surface	Candidate	Loss Profile	MAE [deg]	RMSE [deg]	Mean [%]
global combined	track2g_curve_aware_raw_centered_shape_global	raw plus centered shape	0.003348	0.003682	7.395
global combined	track2g_curve_aware_raw_offset_global	raw plus offset	0.003459	0.003755	7.630
global combined	track2g_curve_aware_pointwise_control_global	pointwise control	0.003578	0.003900	7.911

The curve-aware loss profiles improve over their own pointwise control in the `Fw` and `global` surfaces, but not in the `Bw` surface. They do not overtake the accepted direction-parallel leaders.

Visual Evidence

The `2026-06-09` collage and overlay bundles were regenerated with dedicated Wave 3.3 sections:

- `Forward Wave 3.3 Curve-Aware Training Models` ;
- `Backward Wave 3.3 Curve-Aware Training Models` ;
- `Global Wave 3.3 Curve-Aware Training Models` in the collage report;
- `Forward Reference Tree And Wave 3.3 Overlay` ;
- `Backward Reference Tree And Wave 3.3 Overlay` .

The visual package supports the matrix decision: the curve-aware losses are valid experimental controls, but they do not yet produce the clean curve-shape and offset behavior needed to replace the strongest existing periodic temporal branches.

Campaign Update Ledger

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-06-10	Wave 3.3 curve-aware training refresh	12 pointwise-control, centered-shape, offset, and full-composite <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates	included in the 132 - candidate matrix	dated collage and overlay bundles regenerated with Wave 3.3 sections	verified exploratory baseline; not promoted
2026-06-08	Wave 3.2 harmonic-offset probe refresh	6 clean and harmonic <code>global</code> , <code>Fw</code> , and <code>Bw</code> candidates plus 3 rechecked Wave 3.1 candidates	included in the 120 - candidate matrix	dated collage and overlay bundles regenerated with Wave 3.2 sections	verified exploratory baseline; not promoted
2026-06-04	Wave 3.1 offset-aware probe refresh	3 <code>global</code> , <code>Fw</code> , and <code>Bw</code> sequential residual offset candidates	included in the 114 - candidate matrix	dated collage and overlay bundles regenerated	verified exploratory baseline; not promoted
2026-05-28	Wave 2.3 residual harmonic temporal hybrid refresh	18 <code>global</code> , <code>Fw</code> , and <code>Bw</code> residual harmonic GRU/LSTM candidates	included in the 111 - candidate matrix	collage and overlay reports refreshed with Wave 2.3 sections	verified exploratory baseline; not promoted over Wave 2.2 or accepted TE Curve Verification Pipeline baselines
2026-05-26	Wave 2.2 harmonic temporal hybrid refresh	periodic temporal convolution, GRU, and LSTM <code>global</code> , <code>Fw</code> ,	included	collage and overlay reports refreshed	strongest repository-owned neural branch

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
		and Bw candidates			

Closeout Decision

Wave 3.3 does not change the accepted TE Curve Verification Pipeline baseline. The current direction-parallel decision remains:

- Fw : rcim_retuned_GBM19_Fw remains the strongest overall forward candidate, while track2g_curve_aware_raw_centered_shape_Fw is the strongest Wave 3.3 forward candidate and nearly matches the current repository-owned neural forward leader.
- Bw : periodic_gru_sequence_Bw remains the strongest practical repository-owned backward candidate; the Wave 3.3 backward candidates are weaker than both the current periodic temporal branch and the tree_Bw static baseline.
- global : periodic_gru_sequence_global remains the strongest repository-owned bidirectional neural candidate; within Wave 3.3, the full curve composite loss is the strongest global profile but remains behind the existing periodic temporal global branch. The next modeling step should not continue loss-only tuning as the primary branch. The evidence supports moving to an explicit multi-head shape/offset architecture or a closely related decomposition that keeps harmonic structure, causal runtime inputs, and separate global , Fw , and Bw best-model surfaces.